

Evaluation of a Learned Dirichlet–Neumann Operator and Experiments Conducted 9–16 July 2026

Internal mathematical report

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Abstract

We evaluate a learned Dirichlet–Neumann operator in long water-wave calculations. The learned and reference calculations use the same spatial grid and the same two-stage Gauss–Legendre integrating-factor method. They differ in the evaluation of $q = G(\eta)\xi$: the reference uses a Craig–Sulem expansion through order six, evaluated with eightfold Fourier padding, whereas the learned calculation uses model C21. The order-six calculation is a numerical reference; it is not the exact solution of the continuum problem.

We attempted 2,560 rollouts, consisting of 256 newly generated initial conditions from each of ten panels. Fifty-two reference calculations were excluded because the reference contained a nonfinite value or its relative Hamiltonian drift exceeded 10^{-3} . Among the remaining 2,508 cases, no C21 rollout contained a nonfinite value in η , ξ , or $G_\theta(\eta)\xi$ at any saved time. The empirical median, 95th percentile, 99th percentile, and maximum of the terminal relative elevation error were 0.00563, 0.1693, 0.4854, and 1.3262. Six cases had error above one.

Five of those six cases belong to the Tanaka initial-condition distribution. For these five cases, one optimal periodic translation of the final predicted surface removes 98.14–99.96% of the squared elevation error. The sixth is a Benjamin–Feir case. Its relative error decreases only from 1.1036 to 0.8422 after translation because several important Fourier modes have incorrect amplitudes and mutually inconsistent phase errors.

The report also explains why each experiment during 9–16 July was reasonable before it was run. Earlier models had small errors on reference states but a bad response to perturbations away from those states. This motivated exact evaluation of $G_0 + G_1$, a learned correction that begins at second order in η , self-adjointness in ξ , and a loss based on the Hadamard shape derivative. These changes were made together in C16, after which no nonfinite state was observed on the fixed 320-case test. Because the changes were bundled, this experiment does not identify which ingredient was individually necessary. Later losses aimed at Fourier modes or horizontal wave speed improved average errors but did not improve every large-error case. In two diagnostic Tanaka calculations, none of the tested partial corrections to the DNO error improved both cases. Correcting the complete DNO error reproduced the order-six calculation to approximately 2×10^{-8} . C25 increases the size of the learned correction while retaining the C20 loss. It completed two epochs before stopping during epoch 3 and has no rollout evaluation, so it is not included in the rollout conclusions.

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1 Question addressed by this report

The project has two different numerical questions.

1. Does replacing the numerical Dirichlet–Neumann operator by the learned operator cause the time integration to produce NaN or Inf before the final time?
2. Conditional on the rollout remaining finite, how accurately does the learned calculation reproduce the numerical reference?

These questions must be kept separate. An absence of NaNs is not a statement that the terminal error is small. Conversely, a large but finite phase error does not imply that the time integrator has become unstable.

The principal empirical statement in this report is deliberately narrow:

C21 produced no nonfinite value on the 2,508 reference-valid cases in the new test. It nevertheless produced six finite terminal elevation errors above one.

The following sections define every object in this statement, give the corresponding statistics, and describe the experiments that led to C21.

2 Mathematical and numerical definitions

2.1 Water-wave variables and the Dirichlet–Neumann operator

Let $x \in [0, 2\pi)$ be periodic, let $\eta(x, t)$ be the free-surface elevation, and let $\xi(x, t)$ be the velocity potential restricted to the free surface. For a fluid of constant depth h , the Dirichlet–Neumann operator maps the surface potential to the normal velocity of the surface. We write

$$q(\eta, \xi; h) = G(\eta; h)\xi. \quad (1)$$

In Zakharov variables, with gravity $g = 1$, the equations used by the code are

$$\eta_t = q, \quad (2)$$

$$\xi_t = -\eta - \frac{1}{2}\xi_x^2 + \frac{1}{2} \frac{(q + \eta_x \xi_x)^2}{1 + \eta_x^2}. \quad (3)$$

Thus an error in q enters both equations: it is the complete right-hand side of (2) and appears quadratically in (3).

For the flat surface, the Fourier symbol of the DNO is

$$\widehat{G_0(h)\xi_k} = |k| \tanh(h|k|) \widehat{\xi_k}. \quad (4)$$

The first shape-dependent Craig–Sulem term is evaluated from

$$G_1(\eta; h)\xi = -G_0(h)(\eta G_0(h)\xi) - \partial_x(\eta \partial_x \xi). \quad (5)$$

The cancellation between the two terms in (5) is important for same-sign high-frequency interactions. Asking a neural network to learn the two large terms separately can destroy this cancellation even when its mean value error is small.

2.2 The numerical reference and the learned calculation

All reported calculations use $N_x = 1024$ equispaced points on $[0, 2\pi)$. The numerical reference evaluates

$$G_6(\eta; h)\xi = \sum_{m=0}^6 G_m(\eta; h)\xi \quad (6)$$

with eightfold Fourier padding for products. The nonlinear terms in (3) are evaluated on a grid with twice as many points and then truncated back to the base grid. We call this calculation the *order-six numerical reference*. The notation G_6 refers to the truncated Craig–Sulem series in (6); it does not assert equality with the continuum DNO.

Models C20, C21, and C25 have the form

$$G_\theta(\eta; h)\xi = G_0(h)\xi + G_1(\eta; h)\xi + R_\theta(\eta, \xi, h), \quad R_\theta = O(\eta^2). \quad (7)$$

The terms G_0 and G_1 are evaluated from their formulas, without an additional learned cutoff. The network approximates the remainder beginning with G_2 .

Each learned block is a sum of multiplier sandwiches of the form

$$R_b(\eta, \xi, h) = \sum_{j=1}^r M_{b,j}(h, D) [\phi_{b,j}(\eta, h) M_{b,j}(h, D)\xi]. \quad (8)$$

The same real Fourier multiplier $M_{b,j}$ is used on the two sides of the pointwise multiplication. Consequently (8) is self-adjoint as an operator in ξ for fixed η and h . The feature network has no bias, hence is $O(\eta)$ near $\eta = 0$; its final lift is $f \mapsto f \tanh f = O(f^2)$. Therefore the learned correction in (7) is $O(\eta^2)$ by construction. The sum over blocks and branches is divided by the square root of their number. This normalization controls the size of a parameter update as the number of branches is changed; it does not reduce the set of functions representable after training.

2.3 Time integration

The linear system containing G_0 is solved exactly in Fourier space inside an integrating-factor formulation. The remaining system is advanced by the two-stage Gauss–Legendre Runge–Kutta method. If v_n is the integrating-factor state and F denotes its nonlinear right-hand side, the stage equations are

$$V_1 = v_n + \Delta t (a_{11}F(V_1) + a_{12}F(V_2)), \quad (9)$$

$$V_2 = v_n + \Delta t (a_{21}F(V_1) + a_{22}F(V_2)), \quad (10)$$

$$v_{n+1} = v_n + \frac{\Delta t}{2} (F(V_1) + F(V_2)), \quad (11)$$

where

$$a_{11} = a_{22} = \frac{1}{4}, \quad a_{12} = \frac{1}{4} - \frac{\sqrt{3}}{6}, \quad a_{21} = \frac{1}{4} + \frac{\sqrt{3}}{6}. \quad (12)$$

Four Picard iterations are used to approximate the coupled stages. The inner step is $\Delta t = 0.01$ in every panel. Long Tanaka and Benjamin–Feir calculations are saved every 0.8 time units and end at $T = 200$; the other panels are saved every 0.08 and end at $T = 20$. The nonlinear right-hand side and the updated state are projected onto $|k| \leq 128$, corresponding to the configured filter fraction 0.25. No additional runtime damping, cutoff, or fallback was enabled in the C21 evaluation reported here.

The order-six and learned rollouts use the same numerical choices. Their intended difference is the map used to compute q . The reference stage arithmetic is float64. The C21 network evaluates q in float32 and returns it to the float64 integration harness. Both completed trajectories are cast to float32 before archival and before the validity and error calculations below. Thus all reported error statistics include this mixed-precision implementation of the learned operator and are computed from float32 archives.

2.4 An exact decomposition of numerical rollout error

Let $z = (\eta, \xi)$. Denote one complete filtered GL2 step using G_6 by Φ_6 and the corresponding step using G_θ by Φ_θ :

$$z_{n+1}^6 = \Phi_6(z_n^6), \quad z_{n+1}^\theta = \Phi_\theta(z_n^\theta). \quad (13)$$

Adding and subtracting $\Phi_6(z_n^\theta)$ gives the exact identity

$$\begin{aligned} z_{n+1}^\theta - z_{n+1}^6 = & \underbrace{\Phi_\theta(z_n^\theta) - \Phi_6(z_n^\theta)}_{A_n: \text{ difference caused by the learned DNO at the current model state}} \\ & + \underbrace{\Phi_6(z_n^\theta) - \Phi_6(z_n^6)}_{B_n: \text{ propagation of state error already present}}. \end{aligned} \quad (14)$$

No linearization is used in (14). The identity explains the logic of the experiments in this report. Pointwise supervised training tries to make A_n small on states drawn from the reference data. Rollouts also require A_n to remain controlled at states generated by the imperfect model. Even when A_n is small, the nonlinear reference map can amplify accumulated state error through B_n . Conversely, observing growth in B_n does not show that the numerical reference is wrong: B_n is the response of the same reference map to two different states.

3 Evaluation population and error measures

3.1 Initial-condition panels

The evaluation contains ten named panels, with 256 independently generated initial conditions in each panel. The labels *g0*, *g1*, and *modal* do not denote different equations or different DNO orders in this evaluation. They denote independent random-number streams from the same Tanaka or Benjamin–Feir distribution. Table 1 gives the actual distinctions.

The panels were generated from recorded parameters and the master seed 20,260,715. Each panel uses a separate NumPy `SeedSequence` stream that is absent from the training metadata. The builder checks hashes of the new initial states and rejects duplicates among the new panels. No archived training trajectory was selected. The builder does not, however, compare the hash of every new state against every state in the training archive. Disjointness is therefore established by independent generation and its saved generation record, not by an exhaustive cross-hash. The construction also does not establish generalization to distributions absent from Table 1.

3.2 Reference validity and model finiteness

For a saved numerical trajectory define the discrete Hamiltonian

$$H_j = \frac{\Delta x}{2} \sum_{m=0}^{N_x-1} (\xi_{j,m} q_{j,m} + \eta_{j,m}^2). \quad (15)$$

A reference trajectory is called *valid* if

1. all saved values of η^6 , ξ^6 , and q^6 are finite; and
2. $\max_j |H_j - H_0| / (|H_0| + 10^{-12}) \leq 10^{-3}$.

The threshold was fixed in the evaluation code before the statistics were computed. A learned rollout is called *nonfinite* if any saved value of η^θ , ξ^θ , or q^θ is NaN or Inf. Notice that this definition examines saved

Panels	Construction of one initial condition	Horizon
Tanaka g0, g1	Sum of one to three separated solitary-wave profiles. Total steepness is drawn between .10 and .35. Dirichlet weights are clamped at .05 before a final renormalization to the sampled total; consequently .05 is not a strict lower bound on the final steepness of each crest. Depth is log-uniform on [.01, .30].	$T = 200$
BF g0, g1, modal	A carrier with wavenumber 4–20 and steepness .05–.13, plus two sidebands offset by 2–4 modes with relative amplitudes .05–.20 and independent phases. Second-order corrections are included. Depth is log-uniform on [.5, 4].	$T = 200$
Linear	A nominal single-mode wave with mode 1–20, random direction and phase, and steepness .005–.15. The reference rollout is nevertheless the nonlinear order-six calculation.	$T = 20$
Random sea, deep	Gaussian spectrum with significant height .005–.03, peak wavenumber 2–12, bandwidth 1.5–5, independent phases, and log-uniform depth [5, 25].	$T = 20$
Random sea, finite depth	The same spectral construction with log-uniform depth [.1, 1.5].	$T = 20$
Stokes, deep	A random-phase fifth-order Stokes profile with base mode 1–20, steepness at most .15, and depth in [4, 50] subject to the generator’s kh constraint.	$T = 20$
Stokes, finite depth	A random-phase fifth-order Stokes profile with base mode 14–26, steepness at most .15, and depth in [.02, 1.5] subject to the generator’s kh constraint.	$T = 20$

Table 1: Initial-condition distributions used in the new evaluation. The three Tanaka/BF replicate labels are retained because the corresponding training sources were stored separately; physically, each replicate uses the same distribution as the other panels in its group.

Order-six reference	C21 finite	C21 nonfinite	Total
Valid	2508	0	2508
Invalid	49	3	52
Total	2557	3	2560

Table 2: Reference validity and C21 finiteness. Accuracy statistics below use only the 2,508 reference-valid cases. The three nonfinite C21 calculations on reference-invalid cases are reported here but cannot be used to score accuracy against a valid reference.

frames. It does not separately inspect every intermediate floating-point operation inside a completed GL2 step.

The complete validity table is Among the 52 excluded references, 24 contain a nonfinite archived field and 28 additional finite references exceed the Hamiltonian-drift threshold. The three nonfinite C21 calculations in the second row occur in one linear-panel case and two finite-depth random-sea cases.

3.3 Raw and translation-aligned errors

The primary terminal error is

$$e_\eta(T) = \frac{\|\eta^\theta(\cdot, T) - \eta^6(\cdot, T)\|_{\ell^2}}{\|\eta^6(\cdot, T)\|_{\ell^2}}. \quad (16)$$

To measure how much of this discrepancy is caused only by horizontal wave position, define

$$e_{\eta, \text{shift}}(T) = \min_{s \in [0, 2\pi)} \frac{\|\eta^\theta(\cdot - s, T) - \eta^6(\cdot, T)\|_{\ell^2}}{\|\eta^6(\cdot, T)\|_{\ell^2}}. \quad (17)$$

The shift is applied to the predicted final elevation only. It is not used during the rollout and it is not an acceptance metric. It answers the more limited question: how small would the terminal elevation error be if one were allowed to correct a single periodic translation after the calculation?

The reported fraction of squared elevation error removed by this translation is

$$r_{\text{shift}} = 1 - \left(\frac{e_{\eta, \text{shift}}(T)}{e_\eta(T)} \right)^2. \quad (18)$$

The numerical approximation to the minimization in (17) begins from the eight largest peaks of the discrete circular correlation and refines each candidate by continuous Fourier translation within one grid cell. The smallest resulting value is retained. This is a high-accuracy search over the observed correlation peaks, not a proof that the global continuous minimizer has been found for every profile.

All quantiles below are empirical quantiles of the finite values on the 2,508 reference-valid cases, using NumPy’s linear interpolation convention. They are conditional on the deliberately constructed mixture of panels in Table 1.

4 C21 results on the new initial conditions

4.1 Nonfinite events

No C21 trajectory was nonfinite among the 2,508 reference-valid cases. If one models these cases as independent Bernoulli draws from exactly the same mixture of initial-condition distributions, the one-sided Clopper–Pearson 95% upper confidence bound after zero events in $n = 2508$ trials is

$$p_{\text{upper}} = 1 - 0.05^{1/2508} = 1.194 \times 10^{-3}. \quad (19)$$

This calculation does not prove that the event probability is zero. It also depends on treating the sampled initial conditions as independent draws and does not apply to longer horizons or to a different family mixture.

4.2 Pooled empirical errors

Table 3 reports all 2,508 valid cases as one sample. Half of the cases have raw terminal error below 0.00563, 95% have error below 0.1693, and the largest error is 1.3262. There are 83 errors above 0.25, 23 above 0.50, 11 above 0.75, and 6 above 1.

Error	n	Mean	q_{25}	q_{50}	q_{75}	q_{90}	q_{95}	q_{99}	Max
Raw e_η	2508	.034908	.001229	.005634	.020413	.083930	.169262	.485363	1.326217
After one shift $e_{\eta,\text{shift}}$	2508	.014649	.000631	.003709	.011708	.032467	.058772	.207540	.842246

Table 3: Empirical terminal relative elevation errors for C21. The first row is the primary error (16). The second row applies the diagnostic final-time translation in (17).

Error	Error > .25		Error > .50		Error > .75		Error > 1	
	Count	Rate	Count	Rate	Count	Rate	Count	Rate
Raw e_η	83	3.309%	23	.917%	11	.439%	6	.239%
After one shift $e_{\eta,\text{shift}}$	17	.678%	5	.199%	1	.040%	0	0

Table 4: Strict threshold counts on the same 2,508 cases. The difference between the rows measures the contribution of a single terminal horizontal translation. Seventeen cases still exceed .25 after this correction.

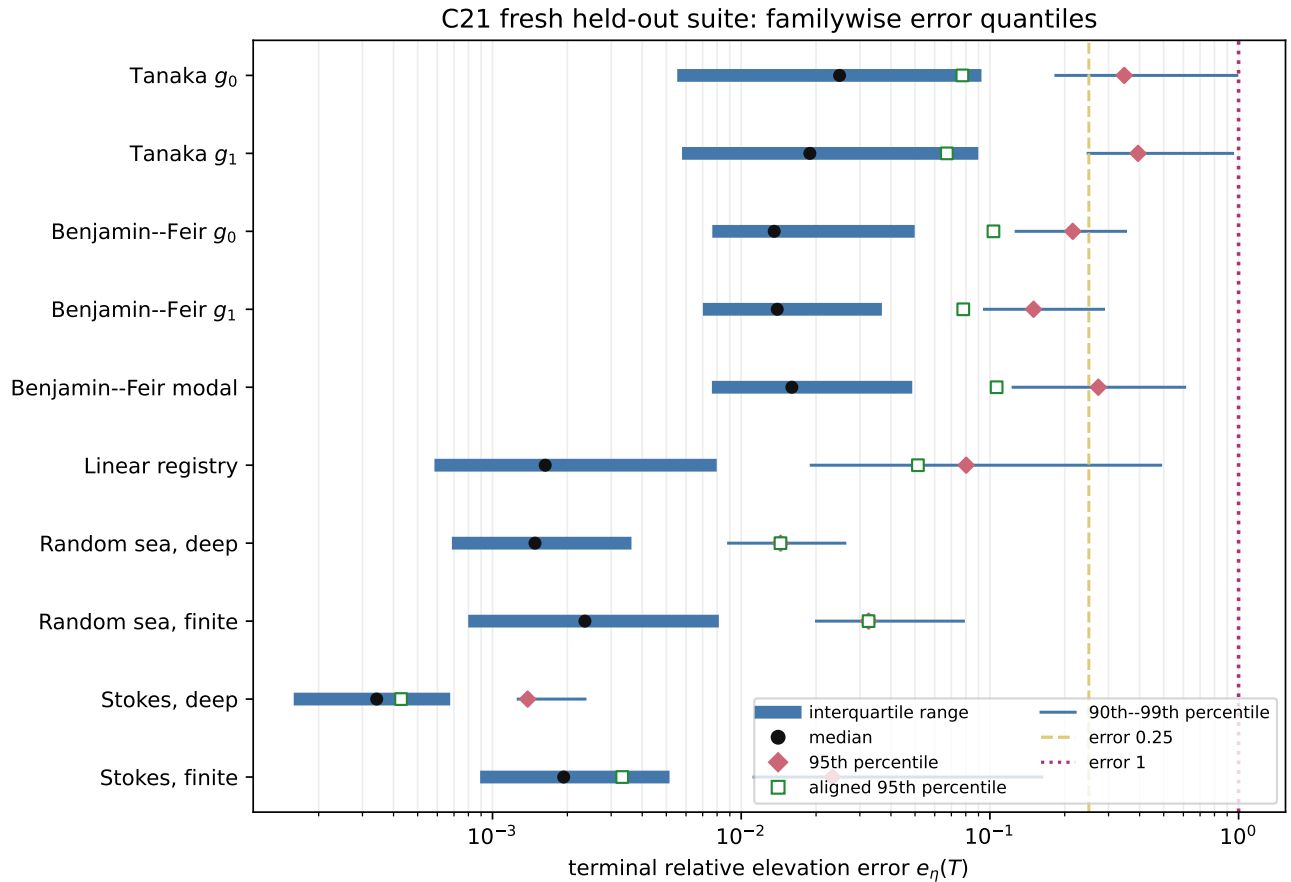


Figure 1: Empirical error quantiles for each initial-condition panel. The black circle is the median raw error. The thick blue interval is the 25th to 75th percentile, the thin blue interval is the 90th to 99th percentile, and the red diamond is the 95th percentile. The open green square is the 95th percentile after the best final-time translation.

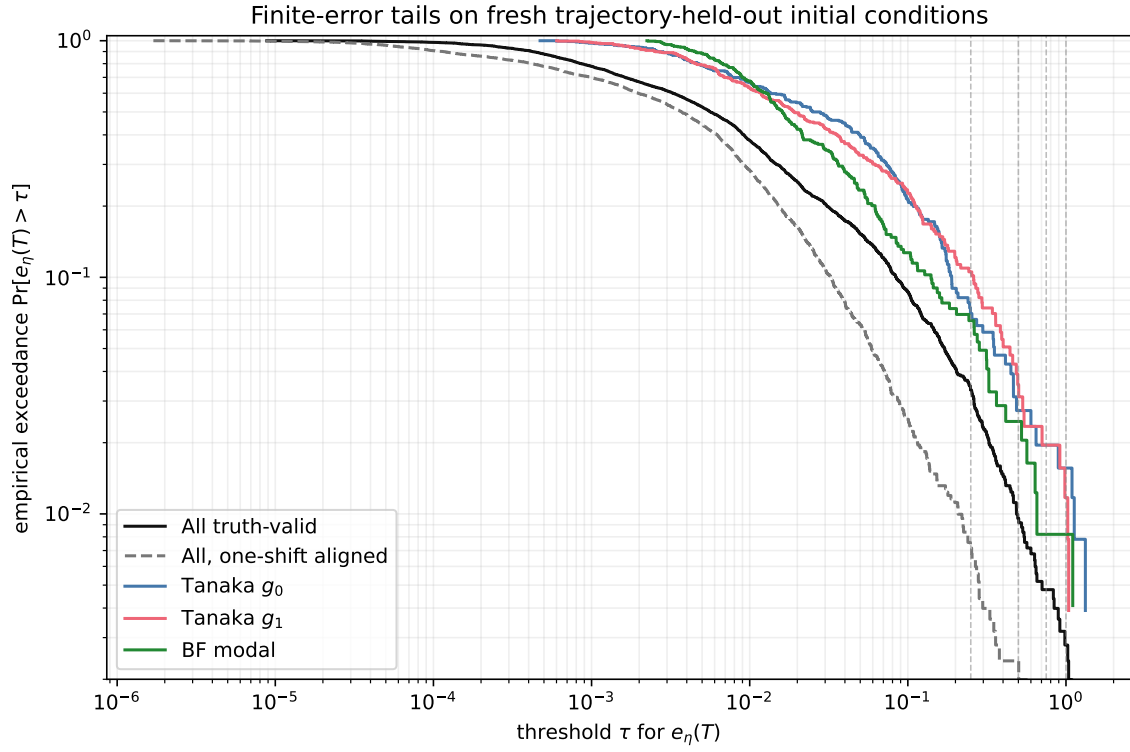


Figure 2: For each error level a , the vertical coordinate is the empirical fraction of valid cases with terminal error larger than a . Solid curves use the raw error. Dashed curves use the error after one final-time translation.

5 Exact training objectives used in the main experiments

This section defines the implemented losses. The names used in the experiment log are only abbreviations for these formulas. None of C16, C20, C21, C22, C23, or C25 uses a Hamiltonian-conservation penalty: its configured coefficient is zero in every one of these runs. The Hamiltonian enters the equations and the reference-validity check, not the training objective.

Let b index a training example, let $N = 1024$, and set

$$q_b^* = G_6(\eta_b)\xi_b, \quad q_b^\theta = G_\theta(\eta_b)\xi_b. \quad (20)$$

Quantities with a tilde are divided by the fixed target scale used in data normalization. Because this is a scalar scale, it cancels from the relative losses below except when a numerical floor is active.

5.1 Implemented one-sided spectral H^1 loss

Let $\mathcal{F}_r$ denote the unweighted real FFT used by JAX and let $\widehat{f}_m^r = (\mathcal{F}_r f)_m$ for $0 \leq m \leq N/2$. The base loss is

$$\ell_{H^1,b} = \frac{\left[\sum_{m=0}^{N/2} (1+m^2) |\widehat{q}_b^\theta - \widehat{q}_{b,m}^*|^2 \right]^{1/2}}{\left[\max \left\{ \sum_{m=0}^{N/2} (1+m^2) |\widehat{q}_{b,m}^*|^2, 10^{-12} \right\} \right]^{1/2}}, \quad L_{H^1} = \frac{1}{B} \sum_b \ell_{H^1,b}. \quad (21)$$

Interior real-FFT modes are not multiplied by two. Thus (21) is the implemented relative one-sided spectral norm, not a claim that a particular quadrature of the continuum H^1 norm was used.

5.2 Soft mode-balanced complex-coefficient loss

This loss uses the forward-normalized real FFT. For $1 \leq k \leq 128$, define

$$\omega_{b,k}^2 = k \tanh(h_b k), \quad (22)$$

$$P_{b,k} = |\widehat{q}_{b,k}^*|^2 + \omega_{b,k}^2 |\widehat{\eta}_{b,k}|^2, \quad (23)$$

$$S_b = \max_{1 \leq k \leq 128} P_{b,k}, \quad (24)$$

$$a_{b,k} = \frac{P_{b,k}}{P_{b,k} + 10^{-4} S_b + 10^{-24}}, \quad (25)$$

$$r_{b,k} = \frac{|\widehat{q}_{b,k}^\theta - \widehat{q}_{b,k}^*|^2}{P_{b,k} + 10^{-6} S_b + 10^{-24}}. \quad (26)$$

The target-dependent quantities P , S , and a are held fixed under differentiation. Set

$$c(r) = \min(r, 100), \quad \rho(r) = \frac{2c(r)}{\sqrt{1+c(r)}+1}. \quad (27)$$

Then

$$\ell_{\text{mode},b} = \frac{\sum_{k=1}^{128} a_{b,k} \rho(r_{b,k})}{\max\{\sum_{k=1}^{128} a_{b,k}, 10^{-24}\}}, \quad L_{\text{mode}} = \frac{1}{B} \sum_b \ell_{\text{mode},b}. \quad (28)$$

This is not equal weighting of a hard set of active modes. The weights change smoothly with target modal energy. Because the complex difference appears in (26), both amplitude and phase errors contribute.

5.3 Gaussian-localized translation-speed loss

First remove the spatial mean from q_b^θ and q_b^* separately, and write their centered difference as $e_b(x)$. Let A_{h_b} be periodic Gaussian smoothing with Fourier multiplier

$$\widehat{A_{h_b} f}_k = e^{-(h_b k)^2/2} \widehat{f}_k. \quad (29)$$

Define

$$C_b(x) = A_{h_b}[\eta_{b,x} e_b](x), \quad (30)$$

$$E_b(x) = \max\{A_{h_b}[\eta_{b,x}^2](x), 0\}, \quad (31)$$

$$E_b^{\max} = \max_x E_b(x), \quad (32)$$

$$\delta c_b(x) = -\frac{C_b(x)}{E_b(x) + 10^{-3} E_b^{\max} + 10^{-30}}, \quad (33)$$

$$w_b(x) = \frac{E_b(x)}{\sum_x E_b(x) + 10^{-30}}. \quad (34)$$

The per-example loss is

$$\ell_{\text{tan},b} = \sum_x w_b(x) \left(\frac{\delta c_b(x)}{\sqrt{h_b}} \right)^2. \quad (35)$$

Only training source identifiers 5, 6, and 14—the two ordinary Tanaka sources and the steep-Tanaka source—are eligible, and an example is omitted if $E_b^{\max} \leq 10^{-20}$. L_{tan} is the mean over eligible examples, with the selected count combined across devices. It is therefore a source-restricted local least-squares translation loss, not a pointwise norm of the complete DNO error.

For a rigid traveling profile $\eta(x, t) = \phi(x - ct)$, one has $q = \eta_t = -c\eta_x$. If the local DNO error is approximately $e_b = -\delta c_b \eta_{b,x}$, then (34) estimates that local speed error. This is the mathematical reason for testing the term. For a deformation error not well represented by η_x , the interpretation is only a projection.

5.4 The C22 γ^2 loss

The C22 quantity is computed by the same local routine. Define

$$\kappa_b^2 = \frac{\sum_x E_b(x)}{\sum_x \eta_b(x)^2 + 10^{-30}}, \quad \ell_{\gamma,b} = h_b \kappa_b^2 \ell_{\text{tan},b}. \quad (36)$$

The elevation in the denominator is not mean-centered. Subject to the local translation approximation $e_b \approx -\delta c_b \eta_{b,x}$ and ignoring the written floors, (36) approximates

$$\frac{\|\delta c_b \eta_{b,x}\|_2^2}{\|\eta_b\|_2^2}, \quad (37)$$

the instantaneous squared relative elevation-error growth caused by a speed error. Equation (37) is not an identity for an arbitrary DNO error. This conditional interpretation is central when C22 is assessed below.

5.5 Randomized finite-secant Hadamard loss

For an arbitrary surface perturbation ζ , the exact DNO satisfies

$$D_\eta G(\eta)[\zeta]\xi + G(\eta)(\zeta B) + \partial_x(\zeta V) = 0, \quad (38)$$

where

$$B = \frac{G(\eta)\xi + \eta_x \xi_x}{1 + \eta_x^2}, \quad V = \xi_x - B\eta_x. \quad (39)$$

For a randomized surface perturbation ζ , the code replaces the derivative in (38) by

$$S_\varepsilon = \frac{G_\theta(\eta + \varepsilon\zeta)\xi - G_\theta(\eta)\xi}{\varepsilon}. \quad (40)$$

The fields B_θ and V_θ are computed from (39) using q_θ , and

$$R_\theta = G_\theta(\eta)(\zeta B_\theta) + \partial_x(\zeta V_\theta). \quad (41)$$

For a field f , define the full-FFT energy

$$\mathcal{E}_{H^1}(f) = N^{-2} \sum_{|k| \leq 128} (1 + k^2) |\widehat{f}_k|^2. \quad (42)$$

The loss is

$$\ell_{\text{Had}} = \frac{\mathcal{E}_{H^1}(S_\varepsilon + R_\theta)}{\mathcal{E}_{H^1}(R_\theta) + 10^{-12}}. \quad (43)$$

The random perturbation begins as Gaussian noise, is restricted to $0 < |k| \leq 128$, divided spectrally by $\sqrt{1 + k^2}$, normalized in physical RMS, and scaled to the larger of the RMS surface variation and 10^{-3} . ε is log-uniform on $[10^{-3}, 3 \times 10^{-3}]$. An active Hadamard batch uses eight examples globally. This is a finite-secant self-consistency loss for G_θ ; a wrong operator can in principle be self-consistent, so supervised anchoring remains necessary.

5.6 C23 eight-step state loss

For C23, a learned path and an order-six path begin at the same training state and take eight GL2 substeps of size .01. Derivatives are propagated through the learned path but not through the order-six path. Both paths use four Picard iterations and the fixed $|k| \leq 128$ projection. At $t_j = j\Delta t$, define for $1 \leq k \leq 128$

$$D_{j,b,k} = |\widehat{\eta^\theta - \eta^6}_{j,b,k}|^2 + G_{0,b,k} |\widehat{\xi^\theta - \xi^6}_{j,b,k}|^2, \quad (44)$$

$$P_{j,b,k} = |\widehat{\eta^6}_{j,b,k}|^2 + G_{0,b,k} |\widehat{\xi^6}_{j,b,k}|^2, \quad (45)$$

$$G_{0,b,k} = k \tanh(h_b k), \quad S_{j,b} = \max_k P_{j,b,k}. \quad (46)$$

With

$$a_{j,b,k} = \frac{P_{j,b,k}}{P_{j,b,k} + 10^{-4} S_{j,b} + 10^{-24}}, \quad r_{j,b,k} = \frac{D_{j,b,k}}{(P_{j,b,k} + 10^{-6} S_{j,b} + 10^{-24}) t_j^2}, \quad (47)$$

the same robust function (27) is applied, followed by the soft modal average and then the mean over eight times and selected examples. We denote the result by L_{FT} . The code also reports an action/phase/polarization decomposition of D , but only the total D enters the optimized loss. The order-six DNO is used only for training supervision; it is not evaluated by C23 at inference.

5.7 Per-step schedules and checkpoint-selection losses

Let s be the optimizer step before an update and $I_{16}(s) = 1$ when s is divisible by 16 and zero otherwise. The exact training objectives are

$$L_{\text{C16},s} = L_{H^1} + I_{16}(s) 0.01 \min(s/500, 1) L_{\text{Had}}, \quad (48)$$

$$L_{C20,s} = L_{H^1} + 6 \min(s/500, 1) L_{\text{mode}} + 10 L_{\text{tan}} + I_{16}(s) 0.01 \min(s/500, 1) L_{\text{Had}}, \quad (49)$$

$$L_{C21,s} = L_{H^1} + 6 \min(s, 1) L_{\text{mode}} + 100 L_{\text{tan}} + I_{16}(s) 0.01 \min(s, 1) L_{\text{Had}}, \quad (50)$$

$$L_{C22,s} = L_{H^1} + 6 \min(s, 1) L_{\text{mode}} + 10 L_{\gamma} + I_{16}(s) 0.01 \min(s, 1) L_{\text{Had}}, \quad (51)$$

$$L_{C23,s} = L_{H^1} + 6 \min(s, 1) L_{\text{mode}} + I_{16}(s) [0.01 \min(s, 1) L_{\text{Had}} + 0.1 \min(s/200, 1) L_{\text{FT},s}], \quad (52)$$

$$L_{C25,s} = L_{C20,s}. \quad (53)$$

C21–C23 reset Adam and the step counter, so their warmups restart at zero. There are 5,964 steps per full epoch at batch size 1024. Thus an every-16 term fires approximately 373 times per epoch; its coefficient must not be interpreted as an every-batch weight. C23 cycles its finite-time microbatch through six source identifiers, so one source is selected approximately once per 96 optimizer steps.

The validation loss used to select checkpoints excludes both the Hadamard and finite-time terms and does not apply training warmup:

Run	Checkpoint-selection loss
C16	L_{H^1}
C20, C25	$L_{H^1} + 6L_{\text{mode}} + 10L_{\text{tan}}$
C21	$L_{H^1} + 6L_{\text{mode}} + 100L_{\text{tan}}$
C22	$L_{H^1} + 6L_{\text{mode}} + 10L_{\gamma}$
C23	$L_{H^1} + 6L_{\text{mode}}$

Table 5: Validation objectives used for checkpoint selection. They differ from the intermittent training objectives in (53).

6 Why the experiments of 9–16 July were worth performing

The experiments were not a sequence of arbitrary loss changes. Each one tested a hypothesis about one of the two terms in the exact error identity (14). This section states the hypothesis before the result, then gives the inference that the result supports. A negative result can be scientifically useful when it distinguishes two plausible causes.

6.1 Hypothesis 1: the NaNs were caused primarily by the time integrator or by aliasing in the nonlinear right-hand side

Reason for the hypothesis. The first nonfinite value was observed during the fixed-point iteration for a GL2 stage. Immediately before failure, intermediate and high Fourier modes grew rapidly. There were therefore at least three plausible numerical causes: (i) the step $\Delta t = .01$ was too large for the nonlinear stage map; (ii) four Picard iterations were insufficient; or (iii) products in (3) aliased unresolved modes back into the retained spectrum. Each mechanism could cause the computed stage map Φ_{θ} to differ from the intended discrete flow even if the supplied DNO value were adequate.

Tests. The saved state immediately before failure was replayed with smaller steps and with the quadratic right-hand side evaluated on a grid twice as large. The same replay was also performed on cases that had remained finite, to check that the modified discretization did not create a new discrepancy in healthy calculations.

Observation. Twofold dealiasing changed healthy trajectories by less than 3.5×10^{-15} over the replay. In failing cases 5, 11, and 22, it delayed the first nonfinite state by only .05, .08, and .05 time units. Step

refinement likewise changed the time at which failure was observed but did not prevent the high-frequency growth from reappearing.

Inference and limitation. These replays are evidence that aliasing and the nonlinear solve amplify a bad state near the end of failure, but are not the first cause in these three cases. They do not prove that the GL2 discretization is accurate relative to the continuum problem. Twofold dealiasing was retained as a numerical correctness improvement because it implements the nonlinear products more faithfully, not because it solved the learned-operator error by itself.

6.2 Hypothesis 2: directly suppressing the growing Fourier band could prevent the nonfinite state

Reason for the hypothesis. In the old failing rollouts, error first increased in modes $32 \leq |k| < 64$ and then spread into $64 \leq |k| \leq 128$. A detector based on the growth of this band identified all then-known Tanaka NaNs without firing on the tested Benjamin–Feir cases. If this growth were not only a precursor but also the main feedback variable, penalizing or damping it could keep the iteration in a region where the stage map remained contractive.

Tests and observations. Several versions were tried.

- C8 penalized the gain of a GL2 stage in modes 16–128. The resulting model still produced NaNs in 3 of 32 Tanaka cases and increased the median Tanaka error from .00907 to .01636.
- C9 increased a soft high-frequency output penalty by a factor of 33. The measured high-frequency diagnostic changed negligibly, and Benjamin–Feir errors increased.
- C10 matched the actual evaluation filter and restricted the stage penalty to modes 64–128. Relative to the previous 3 of 32 count, it reduced the observed Tanaka NaNs to 2 of 32, but did not reach zero and increased several finite errors. Applying an additional output limiter to C10 left the 2 of 32 count unchanged and increased the Benjamin–Feir 95th percentile from .321 to .406.
- C14 imposed a bound on the high-frequency output within the model. Its best finite checkpoint still produced 3 nonfinite trajectories, and 6 of 32 met the historical divergence definition (a terminal elevation error above one or a nonfinite terminal error). Training itself became nonfinite after epoch 25. The proposed eight-block capacity control C15 was stopped before one epoch and supplies no result.

Inference and limitation. The band-growth detector was a useful warning signal, and C10’s partial improvement shows that the growing modes participate in the feedback. However, fixed suppression either left NaNs or reduced accuracy on valid broadband states. These experiments therefore did not justify treating high-frequency content itself as inadmissible. They motivated looking for an earlier error in the DNO map.

6.3 Hypothesis 3: the training data were corrupt or did not constrain the model away from reference trajectories

Reason for the hypothesis. Pointwise supervised training evaluates G_θ on states produced by the reference solver. During a learned rollout, the input state is instead produced by previous applications of G_θ . In terms of (14), training can make A_n small on reference states without controlling A_n on the displaced states actually visited by the model. A label-sign error, nonfinite row, or contaminated set of states sampled shortly before model failure could also give a misleading gradient.

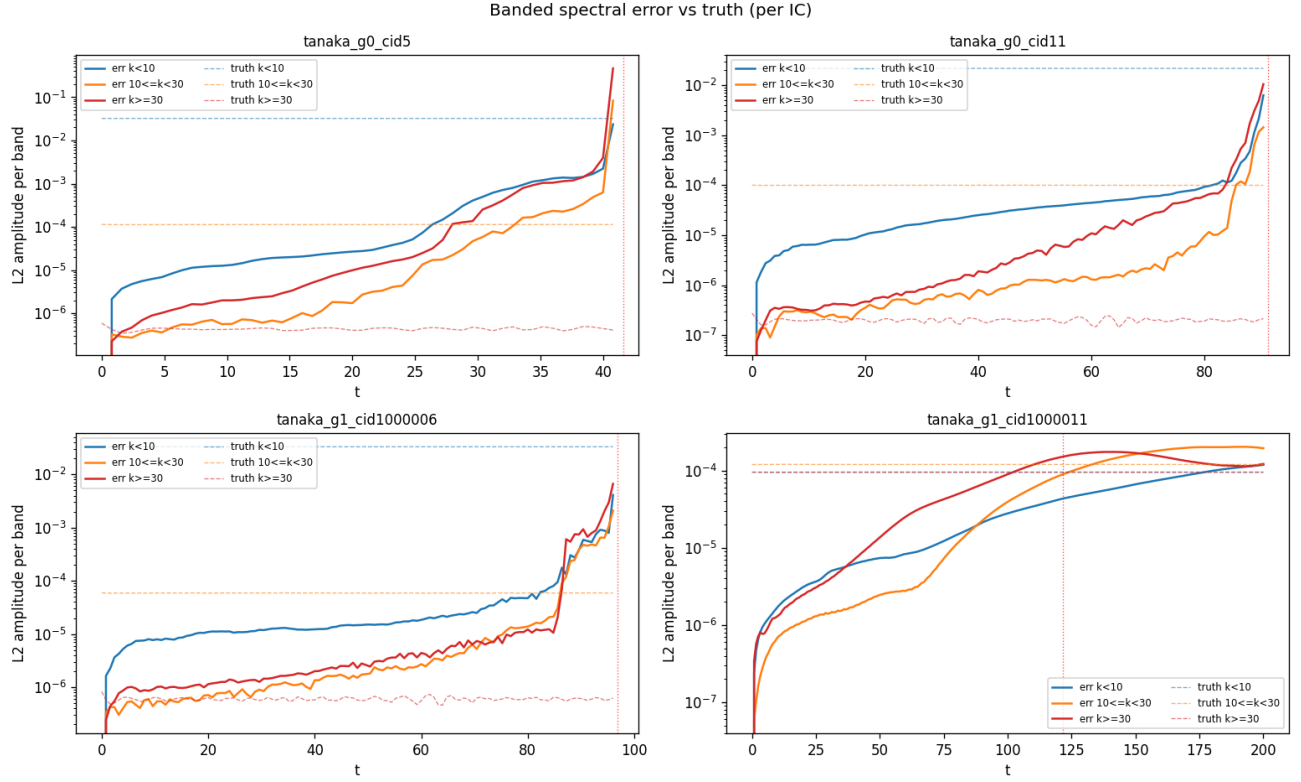


Figure 3: Fourier-band error amplitudes in representative models before C16. This historical figure explains why band-growth penalties and damping were plausible. It is not a figure of the C21 failures described in Section 5.

Tests and observations. A random 5,000-row audit and a separate 512-row source-stratified audit of the base v8/v9 data found no nonfinite η , ξ , or q values. This is a sampled check, not an exhaustive proof that every label is correct. A separate dataset harvested from model trajectories shortly before failure was not clean reference data: 2,943 of 3,855 rows triggered the established high-frequency/sign-change heuristic, and its balanced derivative inherited approximately 43,000 flagged rows out of 100,000. C13 instead generated reference DNO labels near selected difficult Tanaka states and completed training, but its rollout evaluation was canceled. C13 therefore supplies no positive or negative rollout evidence.

Inference and limitation. The audits give no support for widespread nonfinite corruption in the sampled base data. They do show that states harvested from a failing model should not be treated as reference targets merely because they occur near failure. The broader coverage question remains open: even perfectly correct reference snapshots need not cover every direction in which a learned rollout can leave the reference trajectories.

6.4 Hypothesis 4: the model value was accurate on reference states but its derivative with respect to surface shape was wrong

Reason for the hypothesis. For three old failing cases, the relative q error obtained by evaluating the model on the reference state was only about $1.7\text{--}1.9 \times 10^{-4}$ at the examined frames. This appeared too small to explain the rapid subsequent growth by value error alone. We therefore measured the response to a state perturbation. If z^6 is a reference state and δz is the displacement generated by the learned rollout, a finite secant ratio of the form

$$\mathcal{S} = \frac{\|q_\theta(z^6 + \delta z) - q_\theta(z^6)\|_2}{\|\delta z\|_2 + \varepsilon} \quad (54)$$

was approximately 8.1–8.4. The change in model output induced by the displaced state was $2.4\text{--}2.6 \times 10^4$ times the original model bias on the reference state. Thus small pointwise error coexisted with a much worse local response away from the sampled trajectory.

The Hadamard identity (38) constrains the derivative that was not fixed by small value error. On three examined reference states, the learned relative residual in (38) was 1.394, .502, and 2.724; the order-six calculation gave 1.43×10^{-5} , 3.43×10^{-6} , and 3.82×10^{-6} in the same discrete test.

A Fourier perturbation supplied additional evidence. A perturbation inserted at input mode 117 produced an erroneous learned response at output mode 115 of approximately 985.7, compared with 1.47×10^{-5} for the order-six reference. The exact G_1 expression (5) contains the cancellation needed for this same-sign interaction.

Intervention. C16 changed the model and numerical implementation together:

1. $G_0 + G_1$ was evaluated explicitly, with no additional cutoff;
2. the network learned only a correction beginning at $O(\eta^2)$;
3. tied real multipliers made the correction self-adjoint in ξ ;
4. randomized directions ζ were used to penalize the residual of (38); and
5. the quadratic part of (3) was evaluated with twofold Fourier dealiasing.

Observation. The clean epoch-39 C16 calculation produced no nonfinite state in 320 fixed test rollouts without an output cap or runtime damping. Two Tanaka cases had finite raw elevation error above one. By comparison, $G_0 + G_1$ without the learned correction also produced no nonfinite state in 32 Tanaka calculations, but its median and 95th-percentile elevation errors were .207 and .410. Thus the analytic terms alone were finite under this test but too inaccurate.

Inference and limitation. The derivative measurements directly support investigating (38) and the exact G_1 cancellation. The C16 result shows that the complete bundle is compatible with finite long rollouts and useful accuracy. It does not establish that any one of the five changes was individually necessary or sufficient, because they were not ablated one at a time. It also does not prove a zero NaN probability outside the fixed sample.

6.5 Hypothesis 5: the base spectral loss underweighted weak but dynamically important Fourier modes

Reason for the hypothesis. The numerator of (21) is a single sum over modes. A large carrier mode can dominate this sum even when a weak sideband has a large relative complex-coefficient error. Benjamin–Feir evolution depends on energy exchange among a carrier and nearby sidebands, so a loss that is small in aggregate need not resolve that exchange. The normalization in (28) makes a relative error in a weak active mode visible without dividing by an arbitrarily small coefficient.

Intervention. C20 was trained from random initialization for 40 epochs. Relative to C16, it added both $6L_{\text{mode}}$ and $10L_{\text{tan}}$ as shown in (53). The mode term contributed approximately 2.85–2.88% of the final training objective. Because C20 changed two loss terms and retrained from a new initialization, it is a test of the combined objective rather than an ablation of L_{mode} alone.

Observation. All 320 trajectories in the fixed comparison suite remained finite. Relative to C19, the numbers of valid cases with raw terminal error above .25, .50, .75, and 1 changed from 16, 4, 2, and 0 to 10, 1, 1, and 1. The Tanaka-g0 median and 95th percentile improved from .07090 and .37295 to .02281 and .29847; Tanaka-g1 improved from .12107 and .59621 to .05212 and .40361. A fixed Benjamin–Feir case that had error .91198 under C19 decreased to .36379. The one C20 error above one was a finite Tanaka displacement: its raw error was 1.12696, while its error after one translation was .05061.

Inference and limitation. The result justified carrying the mode-balanced term into subsequent models: the combined C20 retraining improved both the Tanaka and selected Benjamin–Feir comparisons while preserving finiteness. It does not establish that mode balance alone caused the improvement. The run also shows why raw and shifted errors must both be reported: better aggregate modal accuracy did not eliminate a slowly accumulated position error.

6.6 Hypothesis 6: horizontal wave-speed error was weakly constrained

Reason for the hypothesis. For a traveling wave $\eta(x, t) = \phi(x - ct)$,

$$q = \eta_t = -c\eta_x. \quad (55)$$

Thus the component of $q^\theta - q^*$ parallel to η_x changes the instantaneous translation speed. A small persistent speed error can produce an $O(1)$ displacement over $T = 200$.

The discrete Hamiltonian (15) cannot by itself detect this error. If $\tau_s f(x) = f(x - s)$, periodicity and translation equivariance of the DNO give

$$H(\tau_s \eta, \tau_s \xi) = H(\eta, \xi). \quad (56)$$

Therefore a prediction may have the correct Hamiltonian and the wrong horizontal position. This is a symmetry statement, independent of whether a Hamiltonian penalty is used; as stated above, these runs did not use one.

For a single coherent profile, the global least-squares speed associated with a DNO value is

$$c(q; \eta) = -\frac{\langle q, \eta_x \rangle}{\|\eta_x\|_2^2 + \varepsilon}. \quad (57)$$

For separated crests, a global inner product can cancel positive error near one crest against negative error near another. This motivated replacing (57) by the localized construction (34)–(35).

Sequence of tests.

1. C18 added a global translation-speed loss to a C16 checkpoint. On the fixed test it improved the Tanaka-g0 median and 95th percentile to .07963 and .48205, but did not improve Tanaka-g1: its 95th percentile increased from .66002 to .93169. Framewise inspection showed cases with different signed errors at different crests, consistent with cancellation in (57).
2. C19 used the Gaussian-localized loss in (35). Under the same guarded fixed-suite protocol, the Tanaka-g0 median and 95th percentile became .07090 and .37295. Several multi-crest cases improved, but the Benjamin–Feir-g1 and linear 95th percentiles increased. Hence local translation speed was a useful target but not the complete DNO error.
3. C21 continued from the C20 best checkpoint for one low-learning-rate epoch, reset Adam, and increased the coefficient of L_{tan} from 10 to 100. The weighted tangent contribution was still only .683% of the mean training objective. On the fixed 64 Tanaka cases, case 27 improved from raw/shifted errors 1.12696/.05061 to .33642/.03050. Case 24 moved in the opposite direction, from .45866/.14579 to .65909/.24546.
4. C22 began from C20, not C21, and replaced L_{tan} by the conditional growth proxy L_γ in (36). The narrow case 27 improved further to .18047/.02688, but case 24 remained .63952/.23662, and the aggregate Tanaka median and 95th percentile were worse than C21.

Inference and limitation. Equations (55) and (56) make the translation-speed hypothesis mathematically natural for traveling profiles, and the experiments confirm that changing the corresponding loss moves the phase-dominated cases. They also show that minimizing the average projected speed error does not bound the worst initial condition or the worst crest. C21 and C22 are one-epoch continuations with reset optimizer state; they test those specific update paths, not a theorem that weights 100 or 10 are optimal. The interpretation of L_γ is conditional on a translation-like DNO error and does not apply to general deformation.

6.7 Matched fixed-panel comparison of C16–C23

Table 6 compares the same 32 Tanaka-g0 and 32 Tanaka-g1 case identifiers. This panel permits casewise comparison among checkpoints but is not trajectory-independent of the old row-wise training split. C19–C23 use the same selective Tanaka runtime rule in this table; C16 does not, so the C16 row is not an exactly matched runtime-rule comparison.

Run	Median	Raw error			Median	After one shift		
		q_{95}	Max	Count > 1		q_{95}	Max	Count > .25
C16	.100430	.602080	1.198696	2	.025218	.187769	.506238	2
C19	.097824	.495824	.792011	0	.013757	.095005	.404157	1
C20	.033030	.358314	1.126959	1	.006689	.057147	.394209	1
C21	.027887	.344140	.659088	0	.005830	.056214	.341471	1
C22	.034449	.376799	.639522	0	.008344	.056436	.361668	1
C23	.033732	.367509	1.141937	1	.006661	.057703	.396699	1

Table 6: Matched 64-case Tanaka comparison. C21 has the smallest median and 95th percentile in this table, but the identity of the worst case changes as the projected losses change.

6.8 Hypothesis 7: an eight-step state loss would better represent nonlinear error propagation

Reason for the hypothesis. The one-state losses in Section 5 compare DNO values at a state generated by the reference solver rather than by previous model steps. The desired quantity is the error after repeated application of Φ_θ . Equation (14) shows that the effect of an early DNO error is propagated through later states. It was therefore reasonable to differentiate through a short segment of the same GL2 map used in evaluation and compare the complete (η, ξ) state rather than only an instantaneous projection of q .

Intervention. C23 used the eight-step loss defined in Section 5, over horizon .08. It was evaluated every 16 optimizer steps on four selected examples and cycled among six source identifiers, so a particular source was used approximately once every 96 steps. It simultaneously set the C21 tangent coefficient from 100 to zero. On batches where it was evaluated, the finite-time term contributed 2.83% of the mean loss; after its 1/16 duty cycle, its approximate epoch-average contribution was .177%.

Observation. Training was finite and the finite-time loss was evaluated on all 373 expected batches. On the fixed 64-case Tanaka panel, C23 had raw median and 95th percentile .03373 and .36751, nearly equal to C20’s .03303 and .35831 and worse than C21’s .02789 and .34414. Case 27 returned to raw/shifted errors 1.14194/.05014, close to C20’s 1.12696/.05061.

Inference and limitation. This experiment did not justify replacing the dense C21 tangent term by the tested sparse eight-step term. It does not show that finite-time differentiation is intrinsically ineffective. The run changed two things at once, used only one epoch, covered .08 time units out of a $T = 200$ rollout, and assigned the new term a small epoch-average weight.

6.9 Hypothesis 8: one identifiable component of the DNO error might be sufficient to repair the phase cases

Reason for the hypothesis. The translation diagnostics suggested that a low-dimensional component of the DNO error might dominate. Before training another loss, this can be tested within the numerical model. At a model state z , define

$$d(z) = q_6(z) - q_\theta(z). \quad (58)$$

A reference-assisted stage calculation evaluates $d(z)$ at every GL2 stage and adds a selected component to $q_\theta(z)$. It is not a usable surrogate, because it calls the expensive order-six DNO. It asks whether correcting that selected component is sufficient in the discrete rollout.

For each active Fourier mode, the modal phase projection is

$$\widehat{P_\phi d}_k = i\widehat{\eta}_k \frac{\text{Im}(\widehat{d}_k \widehat{\eta}_k^*)}{|\widehat{\eta}_k|^2}, \quad (59)$$

with zero output below the target-amplitude floor. This is the component of $\widehat{d}_k$ parallel to $i\widehat{\eta}_k$ in the real two-dimensional coefficient plane. Local alternatives use the least-squares projection onto η_x with Gaussian widths h or $4h$. The radial complement is $d - P_\phi d$, and the complete correction is d .

Observation. The same corrections were applied to fixed Tanaka cases 24 and 27:

Stage correction	Case 24		Case 27	
	Raw	Shifted	Raw	Shifted
None (C21)	.659088	.245457	.336422	.030498
$P_\phi d$	.625276	.119104	1.024433	.042911
Local width h	.508067	.225268	1.079470	.043482
Local width $4h$	.474397	.151743	.134637	.024499
$d - P_\phi d$	.137739	.136753	.535542	.027532
Complete d	2.25×10^{-8}	2.25×10^{-8}	1.94×10^{-8}	1.84×10^{-8}

Table 7: Reference-assisted GL2 stage corrections. The partial correction with the smallest final raw error is different in cases 24 and 27.

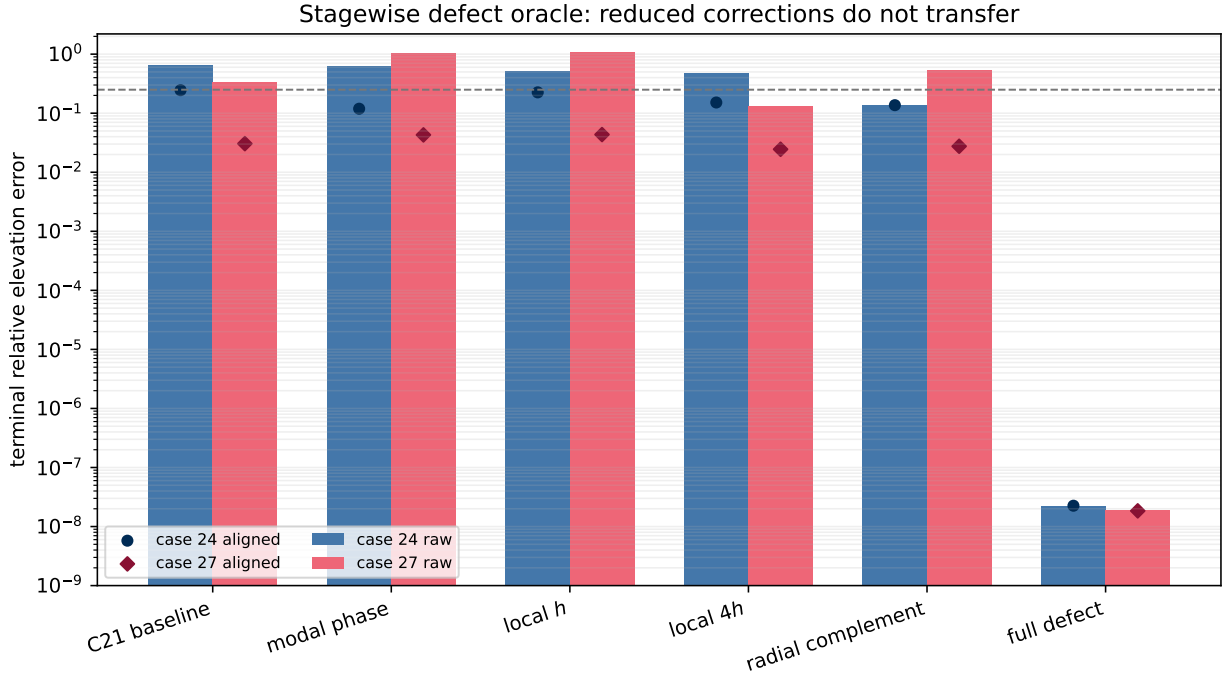


Figure 4: Raw and shifted errors for the corrections in Table 7. Correcting one component can make another case worse. Correcting the complete discrepancy makes the two numerical code paths agree to about 2×10^{-8} .

The local- $4h$ correction gives the smallest partial-correction error for case 27 but reaches raw error .282 at $t = 120$ before decreasing to .135 at $t = 200$. Its terminal improvement therefore includes

finite-time cancellation. The radial complement gives the smallest partial-correction error for case 24 and worsens case 27. The complete correction makes the supplied DNO value equal to $q_6(z)$ at every stage; the resulting agreement is primarily a check that the two numerical implementations are consistent.

Inference and limitation. None of the tested partial corrections improves both cases 24 and 27. The complete reference DNO correction reproduces the reference calculation in both. This does not prove that no other low-dimensional correction exists, that GL2 is converged to the continuum equations, or that these two cases represent all seven initial-condition distributions. It shows only that the specific modal, localized, and radial projections are not sufficient general targets for these two cases.

7 C21 errors separated by initial-condition panel

Table 8: C21 terminal relative elevation error separated by panel.

Panel	n	Raw error			After one shift			
		Median	q_{95}	q_{99}	Max	Median	q_{95}	Max
Tanaka g0	256	.024864	.346807	.982779	1.326217	.007272	.077505	.621166
Tanaka g1	256	.018883	.394226	.945614	1.039555	.006300	.067140	.502451
BF g0	250	.013576	.215374	.351004	.409575	.011009	.103352	.223607
BF g1	246	.013965	.149814	.286310	.327356	.010258	.078148	.279071
BF modal	244	.015984	.272954	.607009	1.103567	.011789	.106436	.842246
Linear	241	.001629	.080299	.485923	.841318	.001354	.051359	.634983
Random sea, deep	256	.001482	.014397	.026108	.036360	.001482	.014386	.036356
Random sea, finite depth	254	.002353	.032527	.078271	.133546	.002352	.032525	.133498
Stokes, deep	256	.000342	.001384	.002354	.008859	.000082	.000429	.000753
Stokes, finite depth	249	.001933	.023278	.161884	.546789	.000405	.003323	.076945

The large errors are not distributed uniformly across the panels. Tanaka g0 contains 17 cases above .25 and 3 above 1. Tanaka g1 contains 26 cases above .25 and 2 above 1. BF modal contains 15 cases above .25 and 1 above 1. The two random-sea panels and the deep Stokes panel contain no case above .25. Because each panel has at most 256 valid cases, the empirical 95th percentile is determined by only about thirteen observations above it. These results are therefore useful empirical error distributions, not estimates of arbitrarily rare-event probabilities.

8 What causes the six errors above one?

This section uses three separate diagnostics. First, the translation error (17) asks whether one final spatial shift explains the elevation discrepancy. Second, Fourier phase rates test whether the observed shift is accumulated through the kinematic equation $\eta_t = q$. Third, a same-state decomposition of q distinguishes the learned operator error from the response of the numerical reference to an already incorrect state.

8.1 Five Tanaka cases: horizontal displacement accumulates over time

Table 9 lists the five Tanaka cases with raw error above one. A single translation removes at least 98.14% of the squared terminal elevation error in every case. In four cases it removes more than 99.89%. This establishes that horizontal position accounts for most of the terminal ℓ^2 discrepancy. It does not, by itself, prove that the dynamical error at all earlier times lies in a single translation eigenvector.

Panel	Index	Raw error	Shifted error	Shift (grid cells)	r_{shift}
Tanaka g0	91	1.326217	.036765	15.736	99.923%
Tanaka g0	186	1.091708	.034695	10.898	99.899%
Tanaka g0	224	1.128034	.154012	9.065	98.136%
Tanaka g1	88	1.026072	.025568	7.159	99.938%
Tanaka g1	202	1.039555	.021458	7.725	99.957%

Table 9: The five Tanaka cases with raw terminal error above one. The last column is defined in (18).

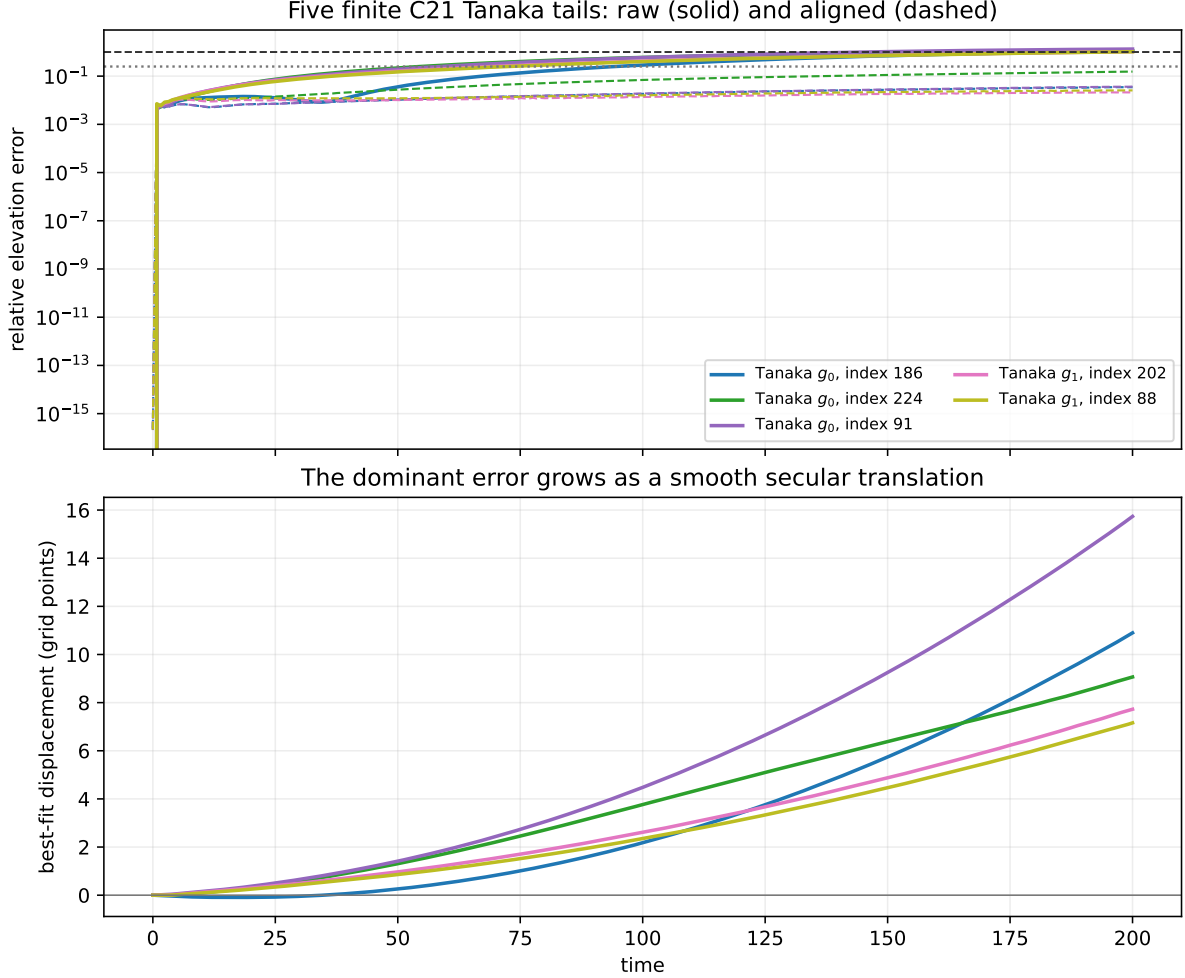


Figure 5: Raw elevation error, error after the best translation at each saved time, and the corresponding translation for the five cases in Table 9. The translations grow over the full interval rather than appearing in one final step.

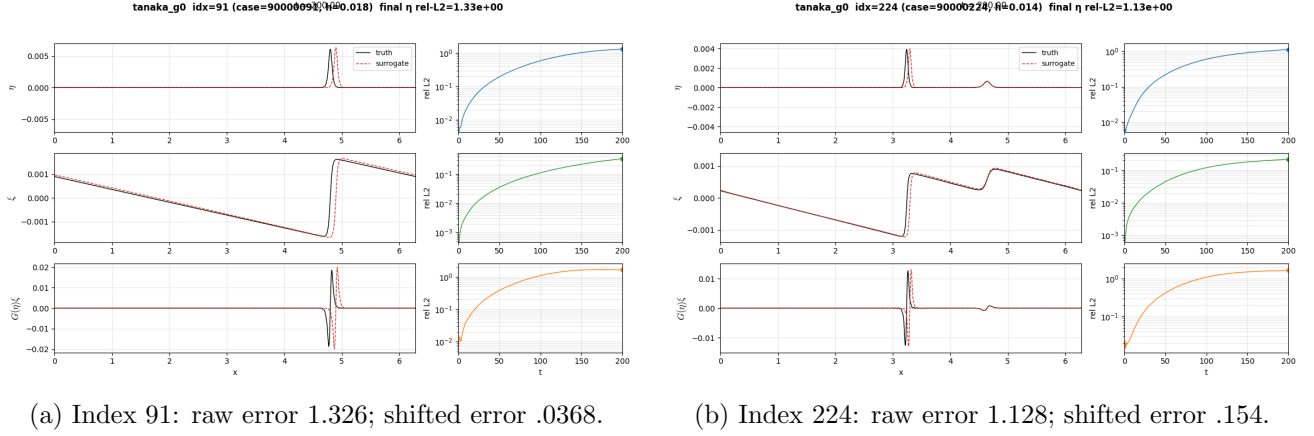


Figure 6: Final order-six elevation (black) and C21 elevation (red dashed). Index 91 is nearly a rigid translation. Index 224 also has a visible shape or relative-crest-position error after the best global shift.

8.1.1 Why an error in q produces a phase error

For a nonzero Fourier coefficient write

$$\hat{\eta}_k(t) = a_k(t)e^{i\phi_k(t)}, \quad a_k(t) > 0. \quad (60)$$

Since $\partial_t \hat{\eta}_k = \hat{q}_k$, division by $\hat{\eta}_k$ gives

$$\frac{\hat{q}_k}{\hat{\eta}_k} = \frac{\dot{a}_k}{a_k} + i\dot{\phi}_k, \quad \dot{\phi}_k = \text{Im} \left(\frac{\hat{q}_k}{\hat{\eta}_k} \right). \quad (61)$$

Equation (61) is an identity whenever $\hat{\eta}_k \neq 0$. In the numerical audit it is applied only to retained modes whose amplitudes exceed the stated activity floor.

For modes 1–8 in the five cases of Table 9, integrating the difference between C21 and reference phase rates reconstructs the measured terminal phase difference to within 3.8×10^{-4} radians. Independently, for modes 1–4, numerical integration of $q = \eta_t$ reconstructs the observed change in $\hat{\eta}_k$ with 1.0–1.7% relative discrepancy. These checks support the following limited conclusion: the saved q discrepancy is consistent with the observed gradual phase accumulation. They do not prove that the continuum solution has the same phase, because the comparison is to the order-six numerical reference.

8.1.2 Learned DNO error versus propagation of existing state error

At any state $z = (\eta, \xi)$ define

$$q_\theta(z) = G_\theta(\eta)\xi, \quad q_6(z) = G_6(\eta)\xi. \quad (62)$$

At the model and reference states, direct addition and subtraction gives

$$\begin{aligned} q_\theta(z_\theta) - q_6(z_6) = & \underbrace{q_\theta(z_\theta) - q_6(z_\theta)}_{\text{DNO approximation error at the model state}} \\ & + \underbrace{q_6(z_\theta) - q_6(z_6)}_{\text{change in the reference DNO caused by state separation}}. \end{aligned} \quad (63)$$

This is the continuous-stage analogue of (14). At the final saved time in the five Tanaka cases, the first term in (63) has relative norm approximately .38–1.19%, whereas the second has relative norm

3.14–11.75%. This does not absolve the learned DNO: the second term is zero when the states agree and grows only after earlier errors have separated the trajectories. It does show why reducing only the instantaneous DNO error on reference-supplied states may not proportionally reduce the final displacement.

8.2 One Benjamin–Feir case: a single translation is insufficient

BF-modal index 22 has raw error 1.10357 and shifted error 0.84225. From (18), one translation removes only 41.75% of the squared error. At Fourier modes 17, 20, and 23, the final predicted-to-reference amplitude ratios are respectively 1.495, .305, and .968. Shifts inferred from the individual phase differences are $-.00699$, $+.07027$, and $-.06199$. Their differing signs show that no single translation can align all three modes. The calculation remains finite; the error is in the relative amplitudes and phases of the carrier and neighboring modes.

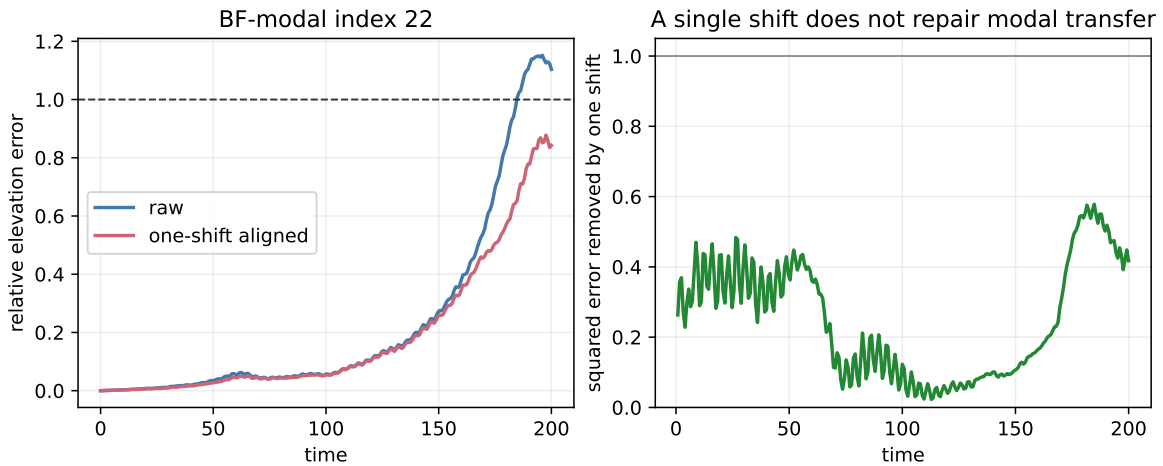


Figure 7: Raw error, error after one horizontal translation, and the fraction of squared error removed for BF-modal index 22. At the final time, the best translation leaves relative elevation error .842.

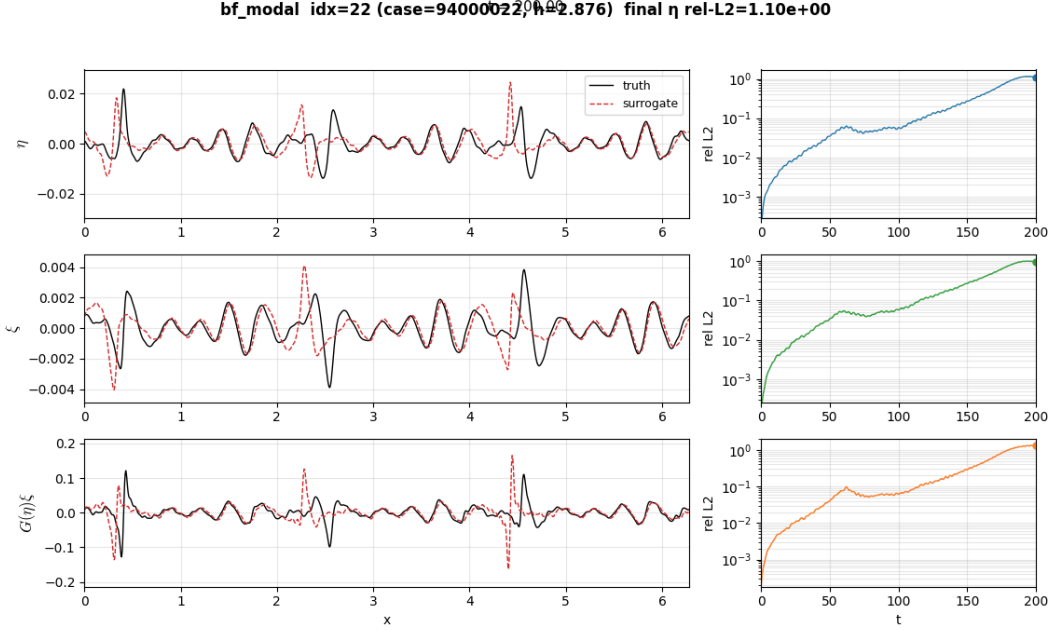


Figure 8: Final order-six and C21 profiles for BF-modal index 22, together with the saved errors in η , ξ , and q . The discrepancy cannot be removed by translating the complete profile.

9 C25: what changed and what has been measured

The reference-assisted calculations in Table 7 did not identify a tested partial correction that improved both cases. Since the complete DNO discrepancy corrected both, one plausible remaining hypothesis is that the learned remainder in (7) lacks either representational or effective optimization capacity. A larger model is worth testing only if the exact $G_0 + G_1$ terms, the $O(\eta^2)$ and self-adjoint construction, the data, and the loss are held fixed.

C25 changes

$$(\text{feature width, branch rank, multiplier hidden width, blocks}) = (512, 256, 128, 8) \longrightarrow (640, 320, 160, 8). \quad (64)$$

The parameter count increases from 860,928 to 1,342,400, or 55.9%. C25 is initialized from random parameters. It retains the exact $G_0 + G_1$ terms, the $O(\eta^2)$ self-adjoint learned remainder, and exactly the C20 objective in (53). No analytic G_m with $m \geq 2$ is added. The γ^2 and finite-time losses are zero.

A 1%-data smoke test completed all 59 batches with finite losses and evaluated the Hadamard term on the expected four batches. In the full-data run, the first two completed epochs are reported in Table 10.

Epoch	Training loss	Validation loss	Validation H^1	Validation mode loss
1	.0114936	.00969421	.00900150	.000106919
2	.00932340	.00874953	.00829122	.0000722955

Table 10: Completed C25 training epochs available at the revision cutoff. The validation loss is defined by Table 5.

Every recorded scalar in these epochs is finite, and expected and observed Hadamard batches are 373/373 in both. The process stopped during epoch 3 after 2,357 of 5,964 training batches; the available

log contains no completed epoch 3 or rollout evaluation. Therefore Table 10 is an optimization-status report, not evidence about long-time error or NaNs.

The capacity hypothesis is supported only if a completed C25 checkpoint is evaluated using the same comparisons as C21. The required sequence is:

1. complete training without a nonfinite loss or a disagreement in optimizer steps or sparse-loss counts between the two GPU copies;
2. evaluate the fixed 64 Tanaka cases, reporting raw and shifted errors;
3. if that comparison improves, evaluate the ten new 256-case panels;
4. compare the nonfinite count, pooled and panelwise empirical quantiles, and all four threshold counts to Tables 2–8.

Even a positive result would show only that this larger trained model performs better. It would not prove that the smaller architecture cannot represent an equally accurate DNO, because optimization and initialization also differ.

10 Conclusions and their limits

The numerical evidence supports the following statements.

1. **Observed finiteness.** Among 2,508 cases for which the order-six numerical reference satisfies the stated validity test, C21 has no saved NaN or Inf in η , ξ , or q . This is a finite-sample statement for the seven sampled initial-condition distributions and the stated time horizons. It is not a stability theorem.
2. **Finite accuracy.** The empirical raw median and 95th percentile are .00563 and .1693. Six cases have raw terminal error above one. Therefore “no observed NaNs” must not be replaced by “no failures” or “uniformly accurate.”
3. **Tanaka displacement.** In five of the six cases above one, a numerically optimized final periodic translation removes 98.14–99.96% of squared elevation error. Their fitted shifts grow over time, and the integrated phase-rate discrepancy from $q = \eta_t$ agrees with the measured phase accumulation. This supports gradual horizontal displacement as the main source of terminal error in those cases. It does not show that every intermediate error lies in a single neutral eigenmode.
4. **Benjamin–Feir error.** In the remaining case above one, the best translation leaves error .8422. Important Fourier modes have different amplitude ratios and phase shifts of different signs. A single horizontal position coordinate cannot describe this case.
5. **Earlier NaNs.** Before C16, small DNO value error on reference states coexisted with order-one residuals in the Hadamard shape identity and a large response to off-trajectory perturbations. Spectral damping and integration changes modified the terminal behavior but did not prevent the old failures. The bundled C16 change—explicit $G_0 + G_1$, an $O(\eta^2)$ self-adjoint remainder, randomized Hadamard consistency, and twofold dealiasing—was followed by no observed nonfinite state on the fixed suite. Because the ingredients were bundled, their individual necessity is not established.
6. **Projected phase losses.** Translation-speed, localized, and γ^2 losses have a clear mathematical rationale and improve selected phase-dominated cases. They do not bound the worst case: an update that improves case 27 can worsen case 24. The reference-assisted corrections likewise show that none of the tested modal, local, or radial components improves both cases.

7. **Complete reference correction.** Adding the complete discrepancy $q_6(z) - q_\theta(z)$ at every GL2 stage makes two diagnostic rollouts agree with their order-six counterparts to about 2×10^{-8} . This verifies numerical consistency of the two code paths once the DNO discrepancy is removed. It is not a continuum convergence result and is not a usable surrogate because it evaluates G_6 .
8. **C25.** C25 asks whether a larger learned remainder performs better when the C20 equations and objective are unchanged. Two completed epochs are finite, but no C25 rollout result exists at this cutoff. No claim about the capacity hypothesis is presently justified.

A Chronological index of experiments, 9–16 July

This appendix places the grouped argument in calendar order. “Did not meet criterion” means the measured result did not justify using that run as the next checkpoint. “No result” means the necessary evaluation was not completed.

Table 11: Experiment chronology.

Date	Run	Controlled question	Measured result	Conclusion permitted by the result
9 Jul	C8	Does a broad GL2 stage-gain loss on modes 16–128 prevent the observed spectral growth?	Tanaka remained at 3 nonfinite cases out of 32; Tanaka and BF medians increased.	The tested broad stage penalty did not meet the finiteness or accuracy criterion.
9 Jul	C9	Does a 33-fold stronger soft penalty on high-frequency q reduce the measured dangerous output?	The diagnostic was essentially unchanged; Tanaka retained 3/32 nonfinite cases and BF errors increased.	Increasing this soft penalty did not control the measured quantity.
9 Jul	C10	Does matching the evaluation filter and targeting only modes 64–128 improve the stage-gain experiment?	Tanaka nonfinite count decreased from 3/32 to 2/32, but finite errors increased and the target of zero was not reached.	The diagnosed band participates in failure, but the tested loss is not a sufficient remedy.
9 Jul	Data audit	Are nonfinite labels, the old sign-change warning, or model-harvested states corrupting training?	Sampled base rows were finite. The model-harvested pre-failure pack had 2,943 flagged rows out of 3,855.	Do not use the model-harvested pack as reference data. The sampled audit is not an exhaustive proof of base-label correctness.
9 Jul	C13	Can exact reference labels near selected difficult states improve C10?	Training completed with finite loss; rollout evaluation was canceled.	No rollout conclusion.
9 Jul	C14	Does an architectural bound on high-frequency output rule out the old failure?	Best finite checkpoint had 3/32 nonfinite Tanaka trajectories and 6/32 under the legacy divergence definition; training became nonfinite after epoch 25.	The tested bound did not meet the criterion.

Date	Run	Controlled question	Measured result	Conclusion permitted by the result
9 Jul	C15	Was C14's poor result caused by using only one block?	Interrupted before epoch 1; No result. no checkpoint.	
9–11 Jul	C16	Does the bundled DNO structure motivated by G_1 cancellation and the Hadamard identity eliminate the observed non-finite calculations?	The clean fixed suite had 0/320 nonfinite model trajectories; two finite Tanaka errors exceeded one.	The bundle met the fixed-sample finiteness criterion. Individual ingredients are not isolated.
11 Jul	C17	Does explicit nondimensionalization of the learned remainder by depth improve shallow cases?	Smoke validation and a case-27 DNO/speed probe worsened.	The tested depth scaling did not justify a full run.
12 Jul	C18	Does a global traveling-wave speed projection reduce the finite Tanaka position error?	Tanaka-g0 improved; Tanaka-g1 95th percentile increased.	One global speed coordinate is insufficient across the two panels.
12–13 Jul	C19	Does a Gaussian-localized projection prevent cancellation between crests?	Several multi-crest Tanaka cases improved; BF-g1 and linear 95th percentiles increased.	Local speed is useful but does not represent the full error.
13–14 Jul	C20	Does a fresh run with the localized term and soft mode-balanced complex loss improve both Tanaka and sideband-sensitive cases?	Fixed suite had 0/320 nonfinite calculations; aggregate errors improved; one finite translated Tanaka case exceeded one.	The combined objective justified C20 as the comparison checkpoint.
14 Jul	C21	Was C20's localized tangent coefficient too small?	One-epoch continuation improved the fixed aggregate and case 27 but worsened case 24. The new 2,508-valid-case evaluation had no observed nonfinite model state and six finite raw errors above one.	Use C21 for the reported aggregate, but do not call weight 100 a uniform phase solution.
15 Jul	C22	Does weighting local speed error by squared relative elevation sensitivity improve narrow shallow profiles?	Case 27 improved; case 24 and aggregate Tanaka statistics did not beat C21.	The tested conditional proxy helps one case but is not a replacement for C21.
15 Jul	C23	Can an eight-step full-state loss replace C21's dense tangent term?	The one-epoch result returned close to C20; the finite-time term contributed approximately .177% after its sparse schedule.	Reject this exact sparse configuration, not finite-time supervision in general.

Date	Run	Controlled question	Measured result	Conclusion permitted by the result
15–16 Jul	Reference-assisted corrections	Is a modal, local, or radial component of $q_6 - q_\theta$ sufficient in two Tanaka cases?	The best partial component differed between cases; only the complete correction produced both references.	None of the tested partial components is sufficient across both cases.
16 Jul	C25	Does a 55.9% larger learned remainder improve the complete DNO approximation with the C20 objective held fixed?	Two epochs completed; training stopped during epoch 3; no rollout evaluation.	No capacity conclusion.

B Reproducibility record

The main numerical artifacts are:

Experiment record `EXPERIMENTS.md`.

C21 checkpoint and fresh evaluation `outputs/c21_tangent_w100_from_c20_20260714_171442/final_ckpt` and `outputs/c21_tangent_w100_from_c20_20260714_171442/eval_final_heldout_unguarded_n256_20260715_035324`.

Initial-condition panels `data/manuscript_ic_panels_v1_20260715`. The recorded master seed is 20,260,715 and the manifest stores panel hashes and generation metadata.

Statistics `notes/weekly_results_stats_20260716.json`.

Figure and statistic builder `scripts/build_weekly_results_report_figures.py`.

Reference-assisted case 24 and case 27 calculations `outputs/c21_tangent_w100_from_c20_20260714_171442/neutral_multiarm_case1000024_20260716_013336` and `neutral_multiarm_case27_20260716_015100`.

C25 `outputs/c25_capacity125_full_20260716_022603`.

The C21 checkpoint used by the fresh evaluation has recorded SHA-256 `232ac8e03099e39ffd75366e20adfdf92dd45d773de32a7aee8e9da23e6c7174`. The reference protocol records hashes of the time-integrator and DNO-series source files in each summary. A remaining record-keeping limitation is that the learned-rollout summaries record model settings and checkpoint identity but do not record hashes of every evaluator source file.

The figure builder is CPU-only. It checks the 2,560 attempted and 2,508 valid counts, the zero valid-reference nonfinite count, and the raw and shifted threshold counts. Recomputed terminal raw errors agree with the archived metric arrays within 1.36×10^{-7} .